A. **Möbius Function.** Next we examine a function that when convoluted with

retrieves f. That function is

The **Möbius function** μ is defined on positive integers n by

$$\mu(n) = \begin{cases} \end{cases}$$

For example:

$\mu(42) =$

$\mu(60) =$

Let's show that μ is multiplicative.

μ is multiplicative.

Proof. Evidently:

Next let m and n be integers greater than 1 such that:

Then, since $m \perp n$, mn is a product of distinct primes if and only if

Hence mn is square–free if and only if

Case 1. All of m, n, and mn are square–free.

Let r and s be the number of distinct prime factors of m and n respectively.

Then mn has:

$\mu(mn) =$

$\mu(m)\mu(n) =$

Case 2. One of m or n is not square–free, and mn is not square–free.

$\mu(mn) =$

$\mu(m)\mu(n) =$

□

B. **Möbius Inversion.** Now we examine the main reason for the Möbius function.

Theorem. $\mu * \mathbb{1} =$

In summation notation, if n is a positive integer

Proof. First we check that both sides are multiplicative.

The left side $\mu * \mathbb{1}$ is multiplicative because:

We have shown that ϵ is multiplicative, thus the right side is also multiplicative.

Because both functions are multiplicative, they will be identical if and only if

Let prime p and positive integer k be given. Then

$\epsilon(p^k) =$

$(\mu * \mathbb{1})(p^k) =$

□

As a corollary we have the Möbius inversion formula.

Möbius Inversion Formula. Let f be arithmetic. Then:

$(f * \mathbb{1}) * \mu =$

Or, in summation notation, if n is a positive integer, then

Proof. By associativity of convolutions we may move parentheses around

$(f * \mathbb{1}) * \mu =$

and then by commutativity of convolutions

$\cdots =$

then using the theorem at the top of the page

$\cdots =$

and lastly using the unit property of ϵ

$\cdots =$

□

Example 1. Compute the convolutions

$\tau * \mu =$

$\sigma * \mu =$

and write out the corresponding summations.

Solution. We recall that

$\tau =$

$\sigma =$

Thus

$\tau * \mu =$

and then using Möbius inversion $(f * \mathbb{1}) * \mu = f$ we obtain:

$\cdots =$

Likewise

$\tau * \sigma =$

and then using Möbius inversion $(f * \mathbb{1}) * \mu = f$ we obtain:

$\cdots =$

In summation notation, if n is a positive integer, then

Example 2. Let S be the arithmetic function that counts square–free divisors.

For example

$S(1) =$

$S(12) =$

Prove that $S =$

Proof. We begin by showing both sides are multiplicative.

Lemma. S is multiplicative.

Proof. We have already seen $S(1) = 1$.

Now let positive integers $m \perp n$ be given.

The positive divisors d of mn may be uniquely expressed in form

Among these, the square–free ones are precisely those satisfying

Thus the number of square–free divisors of mn equals

Hence: $S(mn) =$

□

Lemma. $|\mu| * \mathbb{1}$ is multiplicative.

Proof. In MyOpenMath, you saw that powers of multiplicative functions are:

Hence

Then $|\mu| * \mathbb{1}$ is also multiplicative as

□

Knowing that S and $|\mu| * \mathbb{1}$ are multiplicative, we need only compare them at

Let prime p and positive integer k be given. Then

$S(p^k) =$

$(|\mu| * \mathbb{1})(p^k) =$

□